

Polynomial Regression

A clear walkthrough of your handwritten notes — from Simple Linear Regression to Multivariate Polynomial Regression

Your notes cover one topic in five connected steps. This guide walks through each step in plain language, then shows you the exact page of your notes that step came from, so you can cross-check symbols and handwriting at a glance.

#	Step	What it's about
1	Why we need Polynomial Regression	Linear line can't fit curved (non-linear) data
2	The Hyperplane idea	What a regression line/surface actually represents
3	Polynomial degree formulas	deg = 0, 1, 2, ... n — one general pattern
4	Choosing the right degree	Degree = 1 (too simple) vs Degree = 15 (too complex)
5	Multiple Polynomial Regression	Same idea, extended to 2+ input features

Where you left off in Krish Naik's course:

You've completed Linear Regression. Polynomial Regression is the very next natural step — it reuses the same β (beta) coefficients and the same idea of minimizing error, just with a curved hypothesis instead of a straight line.

1 · Why We Need Polynomial Regression

Start with what you already know

Linear Regression fits a **straight line** to data:

$$h_{\theta}(x) = \beta_0 + \beta_1 x \quad \text{— Simple Linear Regression (1 input feature)}$$

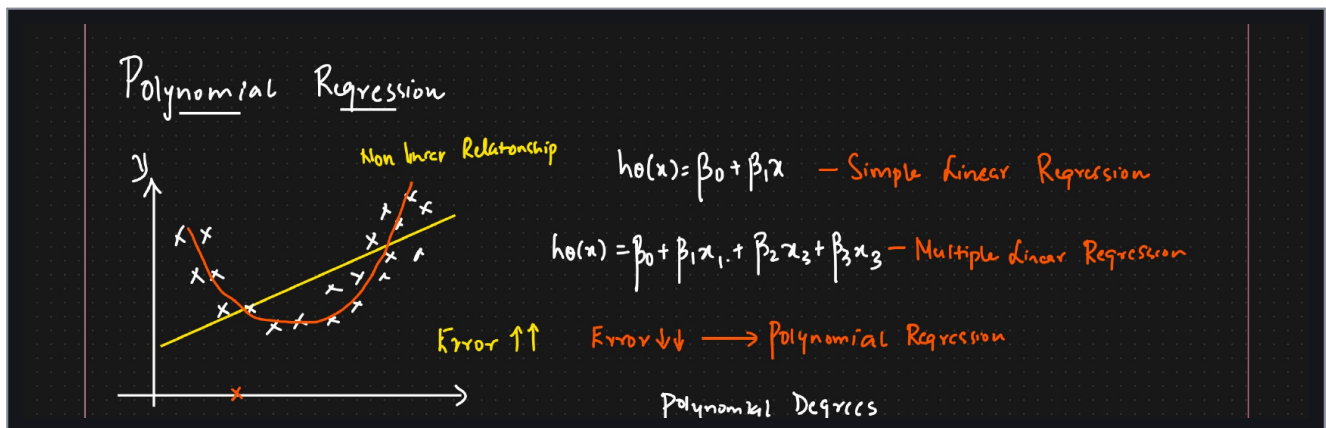
When you have more than one input feature, the same straight-line idea extends into Multiple Linear Regression (still a straight line/plane, just in more dimensions):

$$h_{\theta}(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \quad \text{— Multiple Linear Regression}$$

The problem: not all data is a straight line

Look at the first graph in your notes: the points curve upward like a bowl (a **non-linear relationship**). If you force a straight yellow line through curved data, the line stays far from many points — so the **error is high**. The red curve hugs the points much better, and that curve is what Polynomial Regression produces.

*That's exactly what your notes say in one line: **Error** ↑↑ (straight line) → **Error** ↓↓ (curved line) → use **Polynomial Regression**.*



Your notes — straight line vs. curved fit, and the two base formulas

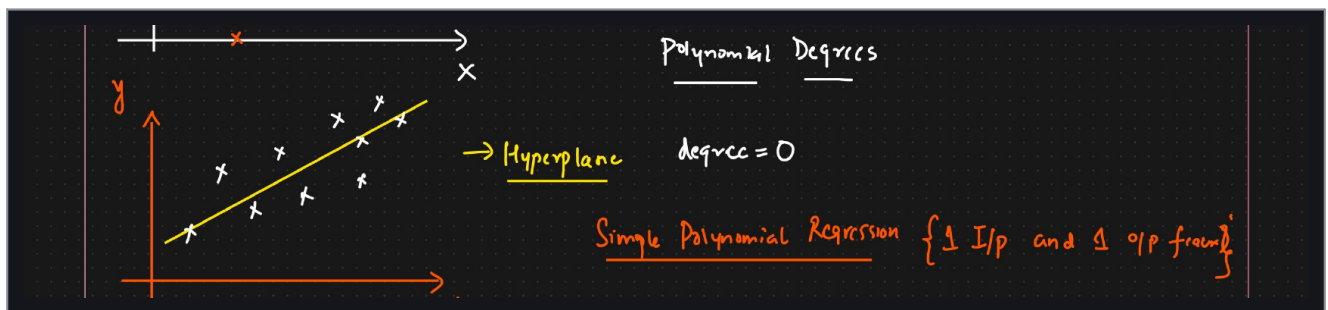
2 · The Hyperplane Idea

What a regression line really is

In the second graph, the yellow line drawn through the scattered points is called a **hyperplane**. In 2 dimensions (1 feature) it's simply a straight line; in 3+ dimensions it becomes a flat plane or surface. Every regression model — linear or polynomial — is really just trying to draw the best-fitting hyperplane through the data.

Simple Polynomial Regression — the setup

Your notes define it precisely: **Simple Polynomial Regression = 1 input feature and 1 output value.** “Simple” here doesn't mean the model is basic — it means there's only one x variable, even though the curve itself can be quite complex. This is the case the next section builds up formula by formula.



Your notes — hyperplane label and the Simple Polynomial Regression definition

3 · Polynomial Degree — Formula by Formula

The **degree** of a polynomial is just “the highest power of x used.” Your notes build this up one degree at a time — each one adds exactly one more $\beta \cdot x^{\text{power}}$ term to the previous formula.

Degree = 0

Constant model

$$h_{\theta}(x) = \beta_0 \cdot x^0$$

$x^0 = 1$ always, so this just outputs a fixed number β_0 for every input — a constant value, no matter what x is.

Degree = 1

→ this IS Simple Linear Regression

$$h_{\theta}(x) = \beta_0 \cdot x^0 + \beta_1 \cdot x^1$$

Since $x^0 = 1$, this simplifies to $\beta_0 + \beta_1 x$ — exactly the straight-line formula from Section 1. So Linear Regression is really just “Polynomial Regression with degree 1.”

Degree = 2

A curve (parabola-like)

$$h_{\theta}(x) = \beta_0 \cdot x^0 + \beta_1 \cdot x^1 + \beta_2 \cdot x^2$$

One more term is added: $\beta_2 x^2$. This lets the hypothesis bend, which is what lets it hug the bowl-shaped data from Section 1 instead of cutting straight through it.

Degree = n (the general pattern)

The full pattern

$$h_{\theta}(x) = \beta_0 x^0 + \beta_1 x^1 + \beta_2 x^2 + \beta_3 x^3 + \dots + \beta_n x^n$$

Every polynomial degree, however high, follows this same repeating pattern — one more $\beta \cdot x^{\text{power}}$ term is added for every increase in degree. Degrees 0, 1 and 2 above are just this same line stopped early.



deg=0

polynomial degree=0

$$h_0(x) = \beta_0 x^0 \Rightarrow \text{Constant value}$$

polynomial degree=1

$$h_1(x) = \beta_0 x^0 + \beta_1 x^{(1)} \rightarrow \text{Simple Linear Regression}$$

polynomial degree=2

$$h_2(x) = \beta_0 x^0 + \beta_1 x^{(1)} + \beta_2 x^{(2)}$$

polynomial degree=n

$$h_n(x) = \beta_0 x^0 + \beta_1 x^{(1)} + \beta_2 x^{(2)} + \beta_3 x^{(3)} + \dots + \beta_n x^{(n)}$$

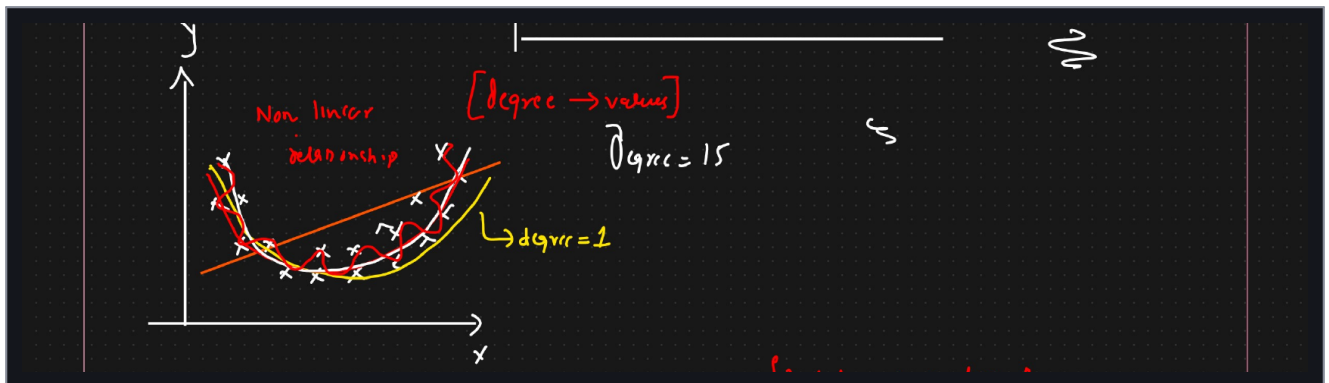
Your notes — degree 0, 1, 2 and the general degree-n formula

4 · Choosing the Right Degree

Degree is a setting *you* choose — it's not learned automatically, and picking it badly causes two opposite problems, both visible in this graph from your notes:

Degree = 1 (yellow line)	Degree = 15 (red wavy curve)
Too simple — stays close to a straight line and misses the bowl shape entirely. This is underfitting : the model is too weak to capture the real pattern.	Too complex — bends so much it wiggles to touch almost every single point. This is overfitting : it memorizes the training points instead of learning the true trend, and will perform badly on new data.

The right degree sits between these two extremes — usually degree 2 or 3 for a mild curve like this one. Your notes mark this as [degree → value]: picking the degree is itself a value you tune, much like choosing k in KNN.



Your notes — degree = 1 underfits, degree = 15 overfits

5 · Multiple Polynomial Regression

Everything so far used just one input feature (x). Your notes end by extending the same idea to **2 independent features** (x_1 and x_2) — this is Multiple Polynomial Regression.

Degree = 1 (still linear)

$$h_{\theta}(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

With degree 1, both features appear only to the power 1 — this is just Multiple Linear Regression again, a flat hyperplane through 2 input dimensions.

Degree = 2

$$h_{\theta}(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2$$

Now each feature also gets its own squared term ($\beta_3 x_1^2$ and $\beta_4 x_2^2$), circled in your notes. Notice the pattern matches Section 3 exactly — it's the same “add one more power” idea, just applied separately to every input feature instead of only one.

The image shows handwritten notes on a blackboard. At the top, there is a small 'x' and a red note '{2 independent features}'. Below this, two equations are written. The first equation is $h_{\theta}(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ with the note 'degree = 1' written to its left. The second equation is $h_{\theta}(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2$ with the note 'degree = 2' written to its left and circled in red. A red arrow points to the x_1 term in the second equation.

Your notes — 2-feature polynomial regression, degree 1 and degree 2

Quick Recap

- Polynomial Regression exists to fix the high error Linear Regression has on curved (non-linear) data.
- Every model — linear or polynomial — is fitting a hyperplane; polynomial terms just let that hyperplane bend.
- Degree 0 = constant, Degree 1 = Linear Regression, Degree 2+ = curves — all one repeating formula pattern.
- Too low a degree underfits; too high (like 15) overfits — the right degree is chosen, not automatic.
- With multiple features, every feature gets its own power terms as the degree increases.